SKYFORCE Interceptor: Sizing Calculations and Final Dimensions Check

Team 08, Anti-Drone Interceptor, Senior Track. Prepared 10 October 2026. Hexarotor, electric, intercept-only mission.

This is one consistent set of numbers. It replaces Sections 1 to 9 of the earlier working document, whose first eight sections used a 22 inch propeller baseline that could not be matched to published motor data. Every figure is tagged: **Published** (manufacturer data), **Computed** (from the equation shown), or **Assumed** (our estimate, still to be verified).

1. What changed from the earlier draft

Earlier claim	Now
U8 II KV100, 447 g, 22 in prop, 9.2 kgf, 41 A, 1965 W	Not backed by published data. Replaced by U8 Lite KV100 with a 28 in prop: 238 g, 7.3 kgf max.
MTOW 18.06 kg, with 15.56 kg of listed parts	Listed parts now sum to 14.78 kg. Sizing uses 18.0 kg. The 3.22 kg gap is shown as unallocated.
18 in prop: 9.3 m/s, 490 W, 2.9 kW	Wrong. Same model gives 8.55 m/s, 387 W, 2.32 kW. Comparison dropped.
At 1500 m thrust falls about 8%	Density falls 13.6%, so thrust at the same rotor speed falls about 13.6%.
Parachute at 5 m/s causes no damage	Now a design target only. 4.74 m/s is a descent rate, not a safety proof.
Reliable net capture at 5 to 15 m, safe motor-failure recovery	Now design targets. Not demonstrated by any calculation here.

2. Design basis

- Sea-level ISA: air density 1.225 kg/m^3, g = 9.81 m/s^2.
- Six rotors in a regular hexagon, 12S battery (44.4 V nominal), figure of merit FM = 0.65 (assumed).
- Mission: about 2 minutes of hover-equivalent flight plus 2 minutes of sprint at 25 m/s, with a 20% energy reserve. A separate tow phase is added for the chosen landing method.
- Drag assumptions (assumed): C_D = 1.2, frontal area 0.15 m^2.

3. Mass ledger

Item	kg	Basis
Airframe: frame, plates, six arm tubes, fasteners	4.50	Assumed. Six 800 mm arm tubes alone are about 1.6 kg at an assumed carbon density of 1600 kg/m^3.
Six motors, U8 Lite KV100	1.43	Published: 238 g each with cable
Six ESCs, Alpha 60A	0.78	Carried over from earlier draft; actual weight not checked
Six propellers, G28x9.2 CF	0.60	Carried over from the 22 in draft; 28 in props are likely heavier
Landing gear	0.80	Assumed
Battery, 12S 8000 mAh	1.97	Assumed: 355 Wh at 180 Wh/kg
Power board, wiring, connectors	0.80	Assumed
Flight controller, GPS, IMU	0.40	Assumed
Companion computer	0.30	Assumed
Camera and gimbal	0.50	Assumed
Datalink radio and antenna	0.50	Assumed
Net launcher frame and mechanism	0.80	Assumed

Item	kg	Basis
Net and tether	0.30	Assumed
CO2 cartridge	0.20	Assumed
Parachute and release	0.40	Assumed
Miscellaneous	0.50	Assumed
Sum of listed items	**14.78**	
Sizing mass used in every calculation below	**18.00**	**Conservative choice**
Unallocated	**3.22**	**Must be assigned to real parts, or the sizing re-run at the lower mass**

The 25 kg takeoff limit is met either way. In the tow phase a captured target of up to 5 kg plus 0.5 kg of net and tether hangs from the drone, giving 23.5 kg.

4. Thrust sizing

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Weight            W = m g = 18.0 x 9.81 = 176.6 N
Max thrust        T_max = 6 x 7.3 kgf x 9.81 = 429.7 N  (7.3 kgf per motor: published)
Thrust-to-weight  T/W = 429.7 / 176.6 = 2.43
Hover thrust      W / 6 = 29.4 N = 3.0 kgf per motor  (41% of max thrust, not a throttle figure)
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Hover thrust being 41% of maximum thrust does not mean the drone hovers at 41% throttle. Throttle-to-thrust behaviour must come from the motor, ESC and propeller data. On the bench data in Section 5, 3.0 kgf sits between the 40% and 50% throttle points.

5. Propulsion selection (verified combination)

Item	Selection	Basis
Motor	T-Motor U8 Lite KV100	Published: 238 g with cable, 12S rated, 27 to 28 in propeller recommended
Propeller	T-Motor G28x9.2 CF (28 in)	Published: the propeller used in the manufacturer test data
ESC	T-Motor Alpha 60A 12S	Published: ESC paired with this motor in the listing. Weight not checked.
Max thrust per motor	7.3 kgf	Published (standard version listing)

Bench data, KV100 + G28x9.2 CF, 12S	Thrust	Current	Power
40% throttle	2.243 kgf	3.5 A	168 W
50% throttle	3.231 kgf	6.0 A	288 W

These are static bench figures that the manufacturer marks as reference only. The earlier 22 in, 9.2 kgf, 41 A figures are withdrawn. A 22 in propeller would need a motor with published thrust data for that size, which was not found.

6. Hover power (two independent estimates)

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Disk area      A = pi R^2 = pi x 0.3556^2 = 0.3973 m^2   (D = 28 in = 0.7112 m)
Induced vel.   v_i = sqrt( T / (2 rho A) ) = sqrt( 29.43 / (2 x 1.225 x 0.3973) ) = 5.50 m/s
Ideal power    P_ideal = T v_i = 29.43 x 5.50 = 161.8 W
Actual power   P = P_ideal / FM = 161.8 / 0.65 = 249 W per motor
Six motors     6 x 249 = 1.49 kW   (Assumed FM)

Bench check    3.0 kgf lies 76.6% of the way from 2.243 to 3.231 kgf
               P = 168 + 0.766 x (288 - 168) = 260 W per motor
               6 x 260 = 1.56 kW   (interpolated bench data)
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The two estimates agree within about 5%. The 1.56 kW bench-based figure is used below.

7. Sprint power and mission energy

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Drag           F_D = 0.5 rho C_D A v^2 = 0.5 x 1.225 x 1.2 x 0.15 x 25^2 = 68.9 N
Body tilt      theta = atan( F_D / W ) = atan( 68.9 / 176.6 ) = 21.3 deg
Parasite pwr   P_par = F_D v = 68.9 x 25 = 1.72 kW
Sprint power   P_sprint = (1.56 + 1.72) x 1.10 = 3.61 kW
Thrust needed  sqrt(W^2 + F_D^2) = sqrt(176.6^2 + 68.9^2) = 189.5 N = 19.3 kgf total
               = 3.2 kgf per motor, against 7.3 kgf available

Sprint energy  3.61 kW x 120 s = 433 kJ = 120.3 Wh
Hover energy   1.56 kW x 120 s = 187 kJ =  52.0 Wh
Mission        172.3 Wh;  with 20% reserve: 206.8 Wh
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The 3.61 kW sprint figure adds hover power to parasite power, which is conservative. It depends on the assumed drag values and does not prove the airframe sustains 25 m/s. Confirm with a propulsion calculator or test, and check battery voltage sag.

8. Landing method and battery size

Chosen method: safe points set as GPS coordinates before launch. After capture the drone checks its position against the safety area. If outside, it tows the netted target to the nearest safe point and releases the parachute there. The tow adds a suspended load for about 60 s (assumed).

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Tow load         18.0 + 5.5 = 23.5 kg;  hover thrust 23.5 / 6 = 3.92 kgf per motor (54% of max)
Tow hover power  1.56 kW x (23.5 / 18.0)^1.5 = 2.33 kW      (power scales as weight^1.5)
Tow energy       2.33 kW x 60 s = 38.8 Wh;  with 20% reserve: 46.5 Wh
Total required   206.8 + 46.5 = 253 Wh
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12S pack	Nominal	Usable at 80%	Intercept only (207 Wh)	With 60 s tow (253 Wh)
6000 mAh	266 Wh	213 Wh	3% margin (marginal)	Not enough
8000 mAh	355 Wh	284 Wh	37% margin	12% margin. Selected.

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Pack current at sprint   3610 W / 44.4 V = 81 A  (10.2 C on 8 Ah)
Current per motor        13.6 A at sprint, against the 60 A ESC rating
Pack current, tow hover  2330 W / 44.4 V = 52 A
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Usable energy is also reduced by ageing, temperature and voltage sag at about 81 A. The pack discharge rating and real voltage under load still need checking.

9. Arm and frame

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Arm length       L >= 1.1 x 711 mm = 782 mm  ->  L = 800 mm
Span             2 x (800 + 355.6) = 2311 mm  (about 2.31 m)

Tube 35 mm OD, 2 mm wall:  I = pi/64 (0.035^4 - 0.031^4) = 2.83e-8 m^4
End load         F = 7.3 x 9.81 + 0.238 x 9.81 = 73.9 N   (max thrust + motor weight)
Root moment      M = F L = 73.9 x 0.800 = 59.1 N m
Bending stress   sigma = M c / I = 59.1 x 0.0175 / 2.83e-8 = 36.5 MPa
Safety factor    400 / 36.5 = 10.9   (400 MPa carbon allowable: assumed)
Tip deflection   delta = F L^3 / (3 E I) = 73.9 x 0.512 / (3 x 70e9 x 2.83e-8) = 6.4 mm
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This is a simplified cantilever check. It does not cover laminate properties, arm-root joints, fasteners, fatigue or impact loads, so 10.9 is not a whole-airframe safety factor. Moving from 22 in to 28 in propellers grows the span from 1.86 m to 2.31 m, versus 1.98 m for the Fortem F700.

10. Parachute and net

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Suspended load   (5.0 + 0.5) x 9.81 = 53.96 N
Canopy area      A = 2 W / (rho C_Dp v^2) = 2 x 53.96 / (1.225 x 0.8 x 5^2) = 4.40 m^2
                 (C_Dp = 0.8, assumed)
Diameter         sqrt(4 A / pi) = 2.37 m  ->  use 2.5 m canopy (A = 4.91 m^2)
Descent rate     v = sqrt( 2 W / (rho C_Dp A) ) = 4.74 m/s
Kinetic energy   0.5 x 5.5 x 4.74^2 = 62 J   (reference only, not an injury threshold)
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Net opening      3 x 1.2 m assumed target span = 3.6 m  ->  4 m
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Opening shock, pendulum motion, deployment reliability, landing surface, net coverage, targeting error and capture probability are not evaluated. The 4 m net and the 5 to 15 m firing range are design targets.

11. Final dimensions

Parameter	Value	Status
Configuration	Hexarotor, six motors in a regular hexagon	Design choice
Sizing mass	18.0 kg (23.5 kg in tow phase)	Computed
Motor / propeller / ESC	U8 Lite KV100 / G28x9.2 CF / Alpha 60A 12S	Published
Propeller diameter	28 in (711 mm)	Published
Arm length (centre to motor)	800 mm	Computed
Arm tube	Carbon, 35 mm OD, 2 mm wall	Computed, simplified check
Tip-to-tip span	2.31 m	Computed
Central body diameter	About 300 mm	Assumed
Height with landing gear	About 450 mm	Assumed
Max thrust / T/W	43.8 kgf / 2.43	Published rating, computed ratio
Hover power	1.56 kW (disk theory 1.49 kW)	Computed
Sprint power at 25 m/s	3.61 kW (conservative)	Computed on assumed drag
Body tilt in sprint	21.3 deg	Computed
Mission energy with reserve	207 Wh intercept only; 253 Wh with tow	Computed
Battery	12S 8000 mAh, 355 Wh nominal	Requirement computed; pack not chosen
Pack current at sprint	81 A	Computed
Arm safety factor / tip deflection	10.9 / 6.4 mm	Simplified cantilever
Parachute canopy	2.5 m diameter, 4.74 m/s descent with 5.5 kg	Computed; canopy not verified
Net	4 m opening, 5 to 15 m firing range	Design target

Check against the problem statement

Requirement	Design	Status
Sprint speed at least 25 m/s	Sprint needs 3.2 of 7.3 kgf per motor	Meets on paper; not demonstrated
Takeoff weight under 25 kg	18.0 kg sizing mass	Met
Net capture, parachute recovery of targets up to 5 kg	4 m net, 2.5 m canopy	Design target; not demonstrated
Detection range at least 1 km, state the sensor	Ground radar, see architecture document	Sensor stated; range from radar data sheet to confirm
Vision or radar terminal guidance	Onboard camera and tracker	Design choice

12. Open items

- Assign the 3.22 kg of unallocated mass to real parts, or re-run the sizing at the lower mass.
- Get the weights of the G28 propeller and the Alpha 60A ESC from the supplier.
- Confirm propulsion performance in a calculator such as eCalc or by test. Bench data is static.
- Replace the assumed drag values (C_D 1.2, 0.15 m^2) with CFD or wind-tunnel results.

- Altitude: at 1500 m, thrust at the same rotor speed is about 13.6% lower, so rotor speed must rise. Check against motor data.
- Motor-failure recovery depends on control authority, not on summed thrust. Not assessed.
- Decide whether the 2.31 m frame is acceptable, or look for a motor with published 22 in thrust data.